BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 137 · Lesson 137 · Third Normal Form: Removing Transitive Dependencies

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a partial functional dependency?

Q2. What does ROM stand for?

Q3. What are the outputs of a half adder when A=1 and B=1?

Q4. What is a composite key?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

3NF — No transitive dependency of non-prime attributes on candidate keys.

transitive dependency — An attribute depends on a key indirectly through another non-key attribute.

functional dependency — Value of one attribute determines another.

update anomaly — Duplicate facts can disagree after an edit.

New learning

Check 2NF first — Remove repeating groups and partial dependencies.

Detect indirect dependency — EmployeeID → DeptID → DeptName; the name depends on department.

Decompose tables — Create Department(DeptID,DeptName) linked from Employee.

Worked example

Employee(EmpID,EmpName,DeptID,DeptName) is in 2NF. Why may it violate 3NF?

1. EmpID determines DeptID — Find the given data and name the required result.

2. DeptID determines name — Explain why this intermediate stage is necessary.

3. Split Department — EmpID identifies employee. DeptName depends on DeptID, not directly on EmpID, producing a transitive dependency. Split Employee(EmpID PK,EmpName,DeptID FK) and Department(DeptID PK,DeptName).

We Do · Practise with guidance

Order(OrderID,CustomerID,CustomerPostcode) stores a customer's postcode on every order. Is a split justified?

Order identifies customer

Find the given data and name the required result.

Customer identifies postcode

Explain why this intermediate stage is necessary.

Split or preserve snapshot

Under the assumption CustomerID determines the customer's current postcode, there is an indirect dependency OrderID→CustomerID→CustomerPostcode. Store Customer(CustomerID PK,Postcode) and Order(OrderID PK,CustomerID FK). Historical shipping addresses may need separate, deliberate snapshots.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Define transitive dependency and give a practical example. [6]

Question 2

Normalise Device(DeviceID,RoomID,RoomName) to 3NF, assuming RoomID determines RoomName. [6]

Question 3

Explain why normalisation can reduce anomalies but may require additional joins. [8]

Exit Ticket · Answer before checking

In 3NF, a non-key value should be copied into every child row merely for convenience.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A non-key attribute depends on only part of a composite candidate key.

Q2. Read-only memory.

Q3. Sum 0, carry 1.

Q4. A key made from two or more attributes together.

You Do pages · Independent questions

Q1. Indirect key→non-key→other non-key chain [3], suitable accurate example [3].

Q2. Device(DeviceID PK,RoomID FK) [3], Room(RoomID PK,RoomName) [3].

Q3. Avoid duplicate facts [2], update/insert/delete benefits [2], joins across tables [2], qualified performance and context [2].

Exit Ticket · Model answer

False. Repeated dependent data can create anomalies.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: